

CompleteMeasFitter incorrectly imported in measurement_error_mitigation

<https://github.com/Qiskit/qiskit/issues/6980>

ROW 36

CompleteMeasFitter incorrectly imported in measurement_error_mitigation.py #6980

New Issue

Closed



omarshehab opened on Sep 3, 2021

Information

- Qiskit Terra version: 0.18.0
- Python version: 3.9.5 (default, May 18 2021, 12:31:01) [Clang 10.0.0]
- Operating system: Darwin

What is the current behavior?

I am currently seeing

```
QiskitError: 'Unknown fitter <qiskit.ignis.mitigation.measurement.fitters.CompleteMeasFitter object at 0x7fcbce6c40>'
```

error, when I pass an `qiskit.ignis.mitigation.measurement.fitters.CompleteMeasFitter` object as a value to the parameter `measurement_error_mitigation_cls` in the constructor of `QuantumInstance` and use the `QuantumInstance` object with `QuantumKernel.evaluate()`.

Steps to reproduce the problem

1. Generate the calibration circuits using `complete_meas_cal()`.
2. Execute the calibration circuits on target hardware (for this week ibm_toronto).
3. Calculate the calibration matrix using `CompleteMeasFitter()`.
4. Pass the calibration matrix calculated in step # 3 as a value for the parameter `measurement_error_mitigation_cls` while creating an instance of `QuantumInstance`. For simplicity, I am not overriding the default values for `cal_matrix_refresh_period` and `measurement_error_mitigation_shots`.
5. Use the `QuantumInstance` object in the standard QSVM pipeline.

What is the expected behavior?

It should not throw any runtime error.

Suggested solutions

`CompleteMeasFitter` incorrectly imported in `measurement_error_mitigation.py`. At line # 141 of `measurement_error_mitigation.py`, `CompleteMeasFitter` should be imported from `qiskit.ignis.mitigation.measurement.fitters` instead of `qiskit.ignis.mitigation.measurement`.



omarshehab added `bug` on Sep 3, 2021



TheGupta2012 on Sep 3, 2021

Contributor

May I work on this @omarshehab ?



Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants





ShellyGarion on Sep 3, 2021

Member ...

I think that the following PR should handle this issue: [#6867](#)
(note that `qiskit-ignis` is going to be deprecated)



TheGupta2012 on Sep 3, 2021

Contributor ...

Okayy 🙌



omarshehab mentioned this on Sep 3, 2021

[Migrate measure mitigation from ignis for QuantumInstance #6867](#)



ShellyGarion on Sep 3, 2021

Member ...

PR [#6867](#) is part of the next release 0.19: <https://github.com/Qiskit/qiskit-terra/milestone/18>



woodsp-ibm on Sep 3, 2021

Member ...

Can you check with the latest version of qiskit - 0.18.1 (you say you have 0.18.0). The issue you describe was fixed

See https://qiskit.org/documentation/release_notes.html#bug-fixes

Fixed an issue where the QuantumInstance class would potentially try to use the CompleteMeasFitter class before it was imported resulting in an error. Fixed [#6774](#)



omarshehab on Sep 4, 2021

Author ...

[@woodsp-ibm](#) , I received the same error with Qiskit Terra 0.18.1.



jakelishman on Sep 4, 2021 - edited by jakelishman

Edits ▾ Member ...

It sounds like you're passing an instance, but the code wants the actual class (i.e. pass `CompleteMeasFitter` directly). The error you reported is consistent with the import having succeeded, but the wrong value having been passed.

(by the way, for reproducibility, it can be a little easier for us if you're able to provide a minimal complete block of code to read and copy/paste, rather than trying to interpret text)



omarshehab on Sep 4, 2021

Author ...

[@jakelishman](#) , I have attached the MWE code here.

[qsvm-readout-error-mitigation-mwe.ipynb.zip](#)





jakelishman on Sep 5, 2021

Member ...

That's not exactly a MWE, but I can see you are setting `measurement_error_mitigation_class` to an *instance* of `CompleteMeasFitter`, but it's meant to be the class (type). You need to pass `CompleteMeasFitter` to construct `QuantumInstance`, not a constructed instance of it.



woodsp-ibm on Sep 5, 2021

Member ...

If an example would be helpful for how to use it see this tutorial https://qiskit.org/documentation/tutorials/algorithms/03_vqe_simulation_with_noise.html#Performance-with-noise-and-measurement-error-mitigation



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omarshehab on Sep 5, 2021

Author ...

@woodsp-ibm, I modified the statement to create `QuantumInstance` as follows.

```
noise_model = NoiseModel.from_backend(backend)
adhoc_backend = QuantumInstance(backend, shots=1024,
                                seed_transpiler=seed, optimization_level = 3, noise_model=noise_model, measure
```



Currently, I am getting the following error.

```
QiskitError: "The noise model is not supported on the selected backend ibmq_quito (<AccountProvider for
```



ShellyGarion on Sep 5, 2021

Member ...

@omarshehab - you cannot have a noise model with ibmq backend, only with Aer simulator:
`backend = Aer.get_backend('aer_simulator')`
the ibmq devices already have noise inside them :)
so, you should either choose an ibmq backend, or Aer simulator with a noise model
please look again at the tutorial suggested above.



jakelishman on Dec 14, 2021

Member ...

I'm closing this now, because I think that if there was a bug, it was fixed, and the other issues were due to problems with the usage. Please feel free to re-open this or another issue if that's not the case.



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jakelishman closed this as completed on Dec 14, 2021